

JAVA METHOD OVERRIDING PROGRAM

Object-Oriented Programming Practical Record

WEEK 7

Question - Method Overriding - Implement RBI as a base class and SBI, ICICI and PNB as derived classes. Demonstrate dynamic polymorphism by displaying the interest rate for the selected bank.

Concept: Inheritance, method overriding and dynamic polymorphism.

Submitted by: G. Pranadeep

Roll No.: A25126552212

Class: 2/4 CSM-D

Branch: Computer Science (Artificial Intelligence / Machine Learning)

AIM

To write, compile and execute the Java program described in the given problem statement.

ALGORITHM

1. Create the RBI base class with `getInterestRate()`.
2. Override `getInterestRate()` in SBI, ICICI and PNB classes.
3. Read a bank name from the user.
4. Store the selected child object in an RBI reference and call `getInterestRate()`.

PROGRAM

```
import java.util.Scanner;

class RBI {
    double getInterestRate() { return 4.0; }
}

class SBI extends RBI {
    @Override double getInterestRate() { return 7.0; }
}

class ICICI extends RBI {
    @Override double getInterestRate() { return 6.5; }
}

class PNB extends RBI {
    @Override double getInterestRate() { return 6.0; }
}

public class Week7BankMethodOverriding {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter bank name (RBI/SBI/ICICI/PNB): ");
        String bankName = scanner.nextLine().trim().toUpperCase();
        RBI bank;
        switch (bankName) {
            case "SBI": bank = new SBI(); break;
            case "ICICI": bank = new ICICI(); break;
            case "PNB": bank = new PNB(); break;
            default: bank = new RBI(); bankName = "RBI";
        }
        System.out.printf("%s rate of interest is: %.1f%%\n", bankName, bank.getInterestRate());
        scanner.close();
    }
}
```

SAMPLE OUTPUT

Enter bank name (RBI/SBI/ICICI/PNB): SBI
SBI rate of interest is: 7.0%

RESULT

Thus, the Java program was written and executed successfully.